

TD2

```
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til\labs> cd ..
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> git branch
* Tom
main
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> pythoon src/main.py
pythoon : Le terme «pythoon» n'est pas reconnu comme nom d'applet de commande, fonction, fichier de script ou
programme exécutable. Vérifiez l'orthographe du nom, ou si un chemin d'accès existe, vérifiez que le chemin d'a
ccès
est correct et réessayez.
Au caractère Ligne:1 : 1
+ pythoon src/main.py
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (pythoon:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> |

PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python src/main.py
=== MarketPulse ===

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Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> cat .\data\sample\instrume
nts.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

```
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> Get-Content data/sample/prices.csv
```

```
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-25,SP500,6713.00,6730.00,6701.00,6715.00,0
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
```

```
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til>
```

```
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> code src/main.py
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python src/main.py
<class 'dict'>
```

```
dict_keys(['instrument', 'benchmark'])
```

```
AAPL
```

```
Apple Inc.
```

```
USD
```

```
SP500
```

```
S&P 500
```

```
=== MarketPulse ===
```

```
Instrument
```

```
AAPL - Apple Inc.
```

```
Last price: 266.20 USD
```

```
Benchmark
```

```
SP500 - S&P 500
```

```
Last level: 6742.00
```

```
Period: 1 month
```

```
Interval: Daily
```

```
Observations
```

```
AAPL: 21
```

```
SP500: 21
```

```
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til>
```

```

2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python src/main.py
<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '
250.00', 'volume': '38000000'}
250.00
<class 'str'>
250.0
<class 'float'>
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> |
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python src/main.py
AAPL first : 250.00
SP500 first : 6600.00
AAPL last : 266.20
SP500 last : 6742.00
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til>

```

```
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python src/main.py
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> |
PS C:\Users\tomda\Documents\1-ESILV\3-A4\v2\S1\Linux\labs\esilv-marketpulse-g07-til> python src/main.py
=== MarketPulse ===

Market configuration
Period : 1 month
Interval : Daily

Instrument
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
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```